

Venkateswaran Subramaniam

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EDUCATION

BITS PILANI

Pilani, Rajasthan, India

MSC IN BIOLOGICAL SCIENCES

Graduated July 2024

CGPA : 8.45/10

First Division

BE IN ELECTRICAL AND

ELECTRONICS ENGINEERING

Graduated July 2024

CGPA : 8.45/10

First Division

COURSEWORK

SOFTWARE

Object Oriented Programming

Data Structures

Programming in Python

C Programming

HARDWARE

Digital VLSI Design

Analog Circuit Design

Embedded Systems Design

Digital Signal Processing

Microcontroller Design/Interfacing

Electronic Circuit Analysis & Design

SELF TAUGHT

Computer Architecture & Verilog

Embedded drivers development

FPGA programming

SKILLS

PROGRAMMING

Experienced:

Python • C • Verilog • SystemVerilog

Familiar:

UVM • Perl • tcl • Xilinx vivado

Version control:

Git • Perforce

TOOLS/APPLICATIONS

Visual Studio • Eclipse • STM32CubeIDE

• MATLAB • Code Composer Studio •

Altium Designer • Keil μ vision 5

EXPERIENCE

RIVOS INC | SoC DESIGN VERIFICATION INTERN

Apr 2025 – Dec 2025 | Bangalore, India

- Updated the SystemVerilog stimulus generator for SoC fabric agent BFM's which helped test coherency and connectivity between CPU and GPU.
- Wrote tests to verify diverse scenarios: Tested DDR coherency scenarios between CPU and GPU; Implemented congestion testing in Last Level Cache (LLC) at GPU side to induce evictions and snoop responses;
- Verified bi-directional CPU-GPU interactions: Validated GPU-to-CPU interrupt generation and confirmed CPU-initiated MMIO register access.
- Tested negative scenarios: data and protocol errors, and other tasks like Dynamic Voltage and Frequency Scaling (DVFS) tests and xprop enablement.
- Enabled toggle/functional coverage flows to find stimulus gaps; ported tests to fullchip models which follow firmware based verification flows, requiring C test programming for fabric and agent init.

CENIZAS LABS | HARDWARE ENGINEER

Jul. 2024 – Apr 2025 | Chennai, India

- R&D and Testing of a low latency (100 μ s) voltage stabilizer for AC applications.
- Led 30+ tests to evaluate buck/boost performance under varying conditions (MOSFETs, diodes, LC size, gate drivers, op-amp drivers, RMS ICs).
- Designed and routed a 4-layer PCB for the TAKTUS balance board, reducing signal integrity issues related to the RF antenna module.
- Developed a firmware abstraction layer for STM32 and TI microcontrollers, including an automated sensor driver development and integration tool that aims to reduce development time by 50%.

NVIDIA GRAPHICS | MEMORY SUBSYSTEM VERIFICATION INTERN

Feb 2024 – Jul 2024 | Bangalore, India

- Implemented low weight coverage runs for the UCF-SoC TB due to which major bugs are caught early, reducing the verification effort by as much as 50%.
- Developed functional coverage automation scripts in python that could generate the coverage class when given a simplified yaml file as an input.
- Monitored coverage reports, reporting any unexpected behaviors in memory transactions and coverage holes related to packet src, dest, transaction type etc.
- Maintained flow related scripts (Python, Perl, tcl) related to the above tasks.

PROJECT

SMART PARKING SYSTEM

Course Project, Fall 2022

- Developed an automated parking system using STM32F4 Microcontroller, MRFC522 RFID Module, and IR sensors to detect vehicle presence and display real-time slot availability on an LCD screen.
- Improved parking efficiency by enabling seamless vehicle identification, reducing driver search time, and providing real-time updates on available slots.